Linux Foundations - Study Notes

Luma University | Learning & Grow | Synthetic demo material

Learning outcomes

Explain file permissions and apply the principle of least privilege.
Inspect listening ports and identify the process that owns a socket.
Use logs to investigate a failed service without exposing credentials.

File permissions

Linux permission bits grant read, write and execute access to the owner, group and others. The command chmod u+x deploy.sh makes a deployment script executable for its owner. Avoid granting write access to every user when a narrower permission is sufficient.

Processes and ports

The command ss -ltnp lists listening TCP sockets and their processes. A service listening on 127.0.0.1 is accessible only from the local host. Bind an application to an appropriate interface only when network access is needed.

Troubleshooting exercise

Start a local API, inspect its listening port, and record the process identifier. Read the service logs, explain a failed startup and write a short recovery checklist. Estimated study time: 35 minutes.

Reflection

Explain why chmod 777 is usually an unsuitable default for a deployment script. Save only sanitized logs in your learning evidence.